

$$= 1.13(3) = 10\%$$

4 C $s_y = \frac{1}{2}gt^2$
 $0.8 = \frac{1}{2}(9.81)t^2$
 $t = 0.40 \text{ s}$

$$v_x = 1.2/0.40 = 3.0 \text{ m s}^{-1}$$

$$v_y = gt = 9.81(0.40) = 3.9 \text{ m s}^{-1}$$

$$v^2 = 3.0^2 + 3.9^2$$

$$v = 4.9 \approx 5 \text{ m s}^{-1}$$

$$\text{OR } \frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mgh$$

$$v^2 = 3.0^2 + 2(9.81)(0.8)$$

$$v = 5 \text{ m s}^{-1}$$

5 C Velocity is positive throughout, except for when it is zero at midpoint. Hence, gradient of